Financial Markets Mastery Roadmap (Beginner → Hedge Fund Level)

Overview

- Level 1 – Market Basics
- Level 2 – Market Microstructure
- Level 3 – Derivatives Pricing
- Level 4 – Volatility Surface
- Level 5 – Quant Trading Models
- Level 6 – Hedge Fund Architecture

Level 1 — Market Basics

- Stocks, bonds, and derivatives
- Exchanges vs OTC markets
- Market participants: retail traders, institutions, market makers
- Indexes and ETFs
- Interest rates and inflation
- Risk vs return
- Portfolio diversification
- Market cycles
- Key ideas: supply & demand, liquidity, bid■ask spread, arbitrage

Level 2 — Market Microstructure

- Order books and price formation
- Limit orders vs market orders
- Liquidity providers and market makers
- High frequency trading
- Dealer hedging
- Slippage and market impact
- Order flow and dark pools
- Important models: Kyle’s Lambda, Glosten–Milgrom

Level 3 — Derivatives Pricing

- Futures and options
- Payoff diagrams
- Arbitrage pricing
- Risk■neutral pricing
- Hedging strategies
- Black■Scholes model variables: stock price, strike price, time, interest rate, volatility
- Greeks: Delta, Gamma, Theta, Vega, Rho
- Concepts: delta hedging, gamma scalping, put■call parity

Level 4 — Volatility Surface

- Implied volatility
- Volatility smile and skew
- Term structure of volatility
- Surface fitting
- Advanced models: Local volatility, Heston model, SABR model
- Identifying mispriced volatility

Level 5 — Quant Trading Models

- Statistical arbitrage
- Mean reversion strategies
- Momentum strategies
- Factor models
- Machine learning in trading
- Monte■Carlo simulation
- Examples: pairs trading, volatility arbitrage
- Tools: Python, C++, R, MATLAB (libraries: numpy, pandas, scipy, QuantLib)

Level 6 — Hedge Fund Architecture

- Market Data Layer
- Data Processing
- Quant Models
- Strategy Engine

- Risk Engine
- Execution Engine
- Exchange connectivity
- Execution algorithms: TWAP, VWAP, smart order routing

Core Skills

- Finance
- Statistics
- Probability
- Programming
- Optimization
- Machine Learning
- Market intuition

Typical Learning Timeline

- Market basics: 1–2 months
- Market microstructure: 2–3 months
- Derivatives pricing: 3–6 months
- Volatility surface: ~6 months
- Quant trading models: ~1 year
- Hedge fund systems: 2+ years